

MODULE LIST & DESCRIPTION SHEET

AlgoNova – AI-Powered DSA & System Design Mastering Platform

The following modules are derived from the current AlgoNova project README and represent the major functional components of the prototype.

Sr.	Module Name	Purpose	Inputs	Outputs	Data Structures Used
1	AI-Powered Interview Simulation	Simulates a strict technical interview with randomized questions, timed assessment, proctoring, coding, spoken explanation and AI evaluation.	Rules, code, answers, explanation, proctoring signals.	Interview score, integrity result, AI feedback and complexity analysis.	Arrays/lists of questions; objects for session state; key-value score and proctoring fields.
2	System Design Sandbox	Interactive environment for designing scalable systems and receiving AI architectural critique.	Design prompt, Excalidraw components and connections.	Architecture grade, bottleneck analysis and actionable feedback.	Graph-like node-edge representation; JSON objects; arrays for components and connections.
3	Resume ATS Screener & Job Matcher	Analyzes resumes using client-side PDF extraction, ATS rules and AI linguistic critique, with job matching.	Resume PDF/text and profile/job skills.	ATS score, keyword analysis, suggestions and matched jobs.	Strings for resume text; arrays/lists for keywords and skills; key-value objects for scores and matches.
4	Interactive DSA Visualizer	Provides step-by-step algorithm visualization and a Bring Your Own Code workflow.	Algorithm selection or custom JavaScript, Python, Java or C++ code.	Execution states, animations, tracing and graphical array visualization.	Arrays for numerical data; execution-state objects; ordered state sequences for step-by-step tracing.
5	Practice HQ – Code Execution & Judging	Provides coding practice, code execution and AI-assisted quality and complexity feedback.	Coding problem, source code and programming language.	Execution result, solution status and optimization feedback.	Objects for problems/submissions; arrays/lists for test cases and results; strings for source code.
6	Gamified Progress & Global Arena	Tracks activity through XP, rank tiers, badges, streaks and a global leaderboard.	Quiz, interview, System Design and resume activity results.	XP, rank, badges, streaks and leaderboard position.	Counters for XP/streaks; arrays/lists for activity and leaderboard entries; objects for badges/ranks.
7	Dual-Layer Authentication & Security	Provides manual and Google OAuth onboarding with JWT authorization and rate limiting.	Credentials or Google OAuth token.	Authenticated session and protected API access.	User objects/records; key-value credential fields; token strings; request/session state.

Data Structures Used – Interpretation

The data-structure column identifies the principal internal data representations associated with each module. These are kept at the conceptual level because the project README describes the module workflows and technologies but does not expose every source-code implementation detail.

Module Interaction Summary

Authentication provides controlled access to the platform. Interview, System Design, DSA visualization, coding practice and resume activities feed the gamified progress layer. The AI layer supports interview evaluation, System Design critique, resume analysis and coding feedback, while MongoDB Atlas provides persistent application data and progress storage.